VIBE CHECK KENYA

AI-POWERED CYBERBULLYING DETECTION

Detecting Harmful social media comments in Kenyan slang, Swahili, and English

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OVERVIEW

Kenya has a fast-growing digital space

Increased internet usage in platforms like

- YouTube
- TikTok
- X
- Facebook
- Instagram

Problem: Increase in cyberbullying and spread of toxicity within Kenyan comment sections

Language Complexity:

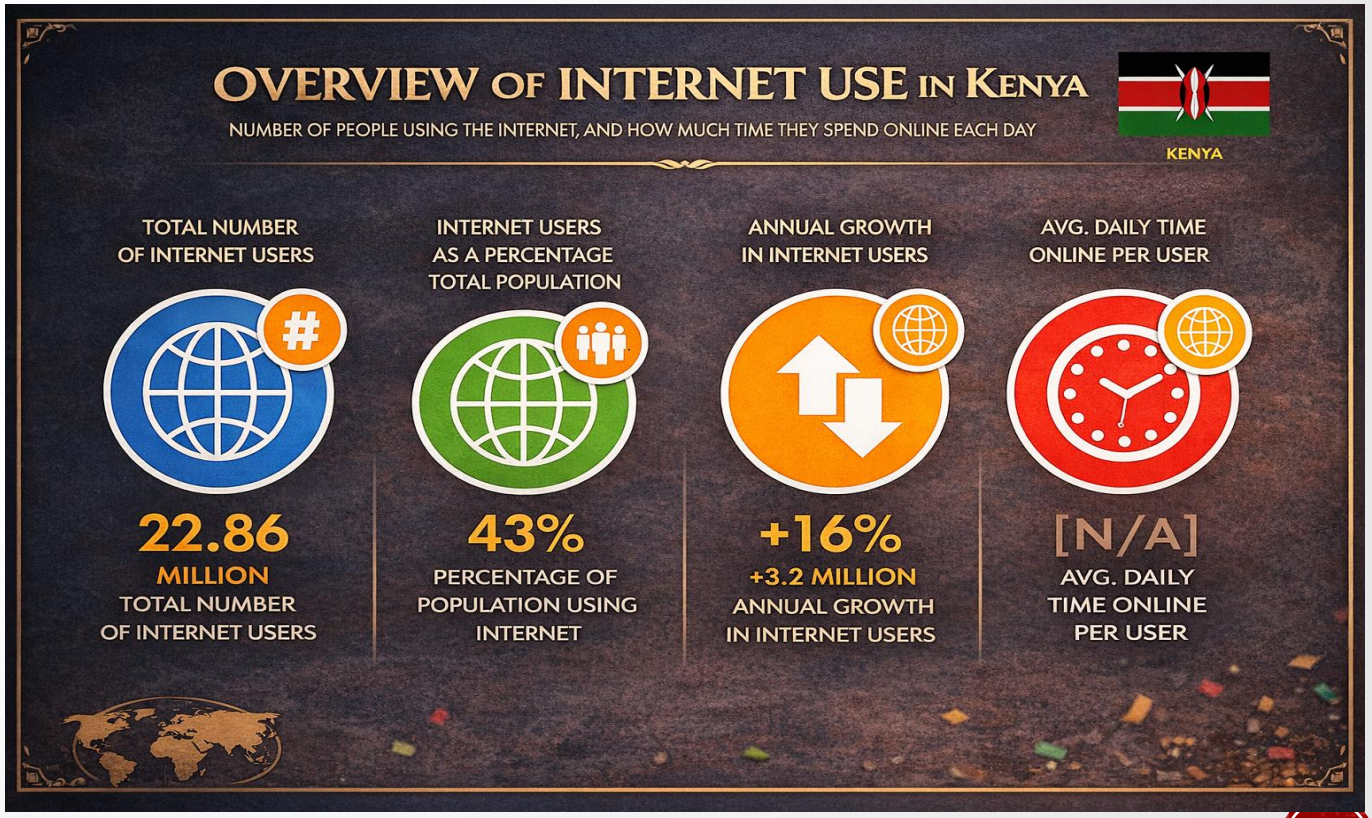
English

Kiswahili

Sheng

Mixed languages(Kenyan Slang)

- With a very High online Kenyan community, most people use Kenyan slang to pass harmful comments, which is often goes undetected in the existing moderation systems



PROBLEM STATEMENT

- Existing Moderation tools are based on Western languages and work mostly in English
- The tools fail to capture linguistic complexity in the Kenyan context
- Fail to detect Sheng and Kenyan slang(code-switching).
- Harmful comments remain online
- Manual moderation is impossible as millions of comments are generated each day.

OBJECTIVES

- Build an AI-powered solution that will detect ;
  - Language of a comment
  - Identify the sentiment(Positive, Neutral/negative)
  - Classify a comment that belongs to which category
  - Detect the subcategory of each comment
  - Create a working prototype

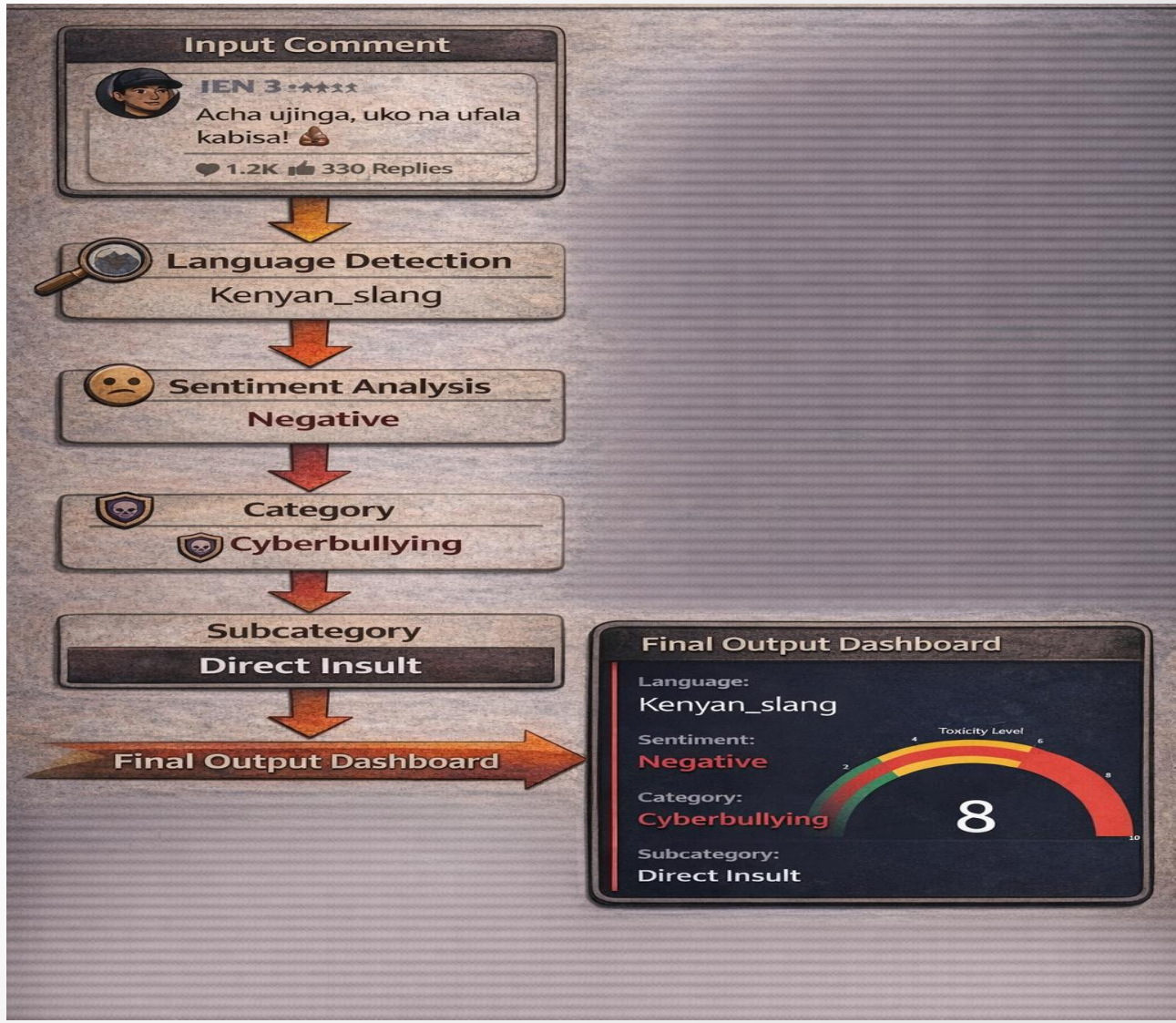
FINDINGS/ANALYSIS

- Collected Data from YouTube(Over 10,000comments) .
- Key Findings:
- Most comments are in Kenyan Slang- mixed languages in a single comment
- The Kenyan comment section is dynamic and cannot be detected by the existing moderation tools.
- Most comments have a neutral sentiment
- Most comments are constructive criticism
- It's difficult to differentiate some harmful comments from those with sarcasm/ irony, which is common in the Kenyan context.
- Few comments contain hate speech, offensive language, and cyberbullying.
- Lack of standard spelling and abbreviations in the Kenyan slang
- Context matter difficult to distinguish categories
- The same word can be harmless or offensive depending on the usage.

MODEL/SOLUTION/OUTPUT

- Build a hierarchical system to identify Kenyan comments
- Step 1: Identify the dominant language of the comment
- Step 2: Detect the sentiment
- Step 3: Classify the major category of the comment
- Step 4: Identify specific subcategory

- OUTPUT



CONCLUSION

- Successfully developed a multilingual cyberbullying detection system.
- Successfully addressed language complexity in Kenyan social media.
- The model captures Kenyan slang, code mixing, and informal expressions used in the Kenyan context.
- The hierarchical approach improved the model accuracy by breaking the problem into smaller, more manageable tasks.
- The system identifies the general and specific behavior in a comment
- Demonstrated high potential for real-time content moderation useful for content creators, bloggers, and social media users

- This system contributes to a safer and more inclusive digital space in Kenya

RECOMMENDATION

- 🌐 Platforms
- 👩‍💻 Creators

- Integrate this system into social media comments.
- A great moderation tool for real-time detection of harmful comments.
- Help content creators automatically filter toxic comments
- This will improve audience engagement by minimizing negativity.

- 🔄 Improvement
- ⚖️ Human + AI

- Regularly update since Sheng and Kenyan slang are dynamic to detect emerging toxic patterns.
- Retrain the model with new changes for better accuracy
- The system should combine AI predictions and human review to minimize false positives.

Policy & Community Use

- Can support:
  - Online safety initiatives
  - Digital literacy programs
- Promote responsible online behavior

FUTURE WORKS

📊 Data	🔄 End-to-End Automated Pipeline	🌐 Scalable Deployment & API Integration	📱 Cross-Platform Integration	📈 Real-Time Monitoring Dashboard	⚖️ Ethical AI & Bias Reduction	🔐 Privacy & Data Protection
• Collect more data from multiple platforms, get more than 500,000 Kenyan comments) • Update sheng slang vocabulary	Develop a fully automated pipeline: • Data collection(from APIs like TikTok, X) • Data cleaning and preprocessing • Feature engineering and model training • Model evaluation and deployment	Develop a REST API for easy integration to external system Enable deployment on cloud platforms Ensure reliability for real-time moderation.	• Extend the system beyond YouTube to other platforms such as TikTok and Facebook • Enable a centralized moderation across platforms	▪ Build an interactive dashboard for : ▪ Live comment analysis ▪ Toxicity trends and insights ▪ Alerts for harmful content spikes ▪ Provide analytics for content creators	Minimize biases related to : Names Language Regional expressions Ensure fairness and inclusivity in predictions.	Ensure user data is anonymized and protected Comply with data protection standards and policies

DEMO SHOWCASING

TEAM CONTRIBUTIONS

- 👩‍💻 **Stella Kiarie – Project Lead & ML Developer**
- 🧠 **Kumati Dapash – Machine Learning Engineer**
- 📊 **Morvine Otieno – Data Analyst**
- 🎨 **Doris Mutie – Deployment & UI Developer**

Any Questions?